

Production of ^{11}C , ^{13}N and ^{15}O in proton-induced nuclear reactions up to 200 MeV

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Production cross sections of light positron emitters ^{11}C ($t_{1/2}=20.36$ min), ^{13}N ($t_{1/2}=9.97$ min) and ^{15}O ($t_{1/2}=122$ s) have been measured for proton-induced reactions on the main relevant constituents of the human body (C, N and O) from threshold up to 200 MeV of proton incident energy. This has been accomplished by means of two complementary experiments at the clinical proton treatment centers WPE (Essen) and HIT (Heidelberg), both in Germany. In the first case, the multi-foil activation technique was combined with PET imaging, a methodology previously developed at CNA (Seville, Spain). In the second experiment, aimed at energies below 50 MeV, the activation of single foils was measured with conventional LaBr₃ detectors. In both cases the IAEA monitor reaction $^{nat}\text{Cu}(p,x)^{63}\text{Zn}$ data was used for validation purposes.

The most complete and accurate set of excitation functions for the production of light-positron emitters ^{11}C , ^{13}N and ^{15}O in proton induced reactions on human tissue up to 200 MeV is presented. These data are needed for nuclear reaction model development and are extremely important to improve the accuracy of the PET range verification in proton radiotherapy. Seven excitation functions have been measured: $^{12}\text{C}(p,x)^{11}\text{C}$, $^{14}\text{N}(p,x)^{13}\text{C}$, $^{14}\text{N}(p,x)^{13}\text{N}$, $^{14}\text{N}(p,\gamma)^{15}\text{O}$, $^{16}\text{O}(p,x)^{11}\text{C}$, $^{16}\text{O}(p,x)^{13}\text{N}$ and $^{16}\text{O}(p,x)^{15}\text{O}$ up to 200 MeV of proton incident energy.

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I. INTRODUCTION AND MOTIVATION

Excitation function of light-positron emitters in proton-induced nuclear reactions are of fundamental interest for in-vivo treatment verification methods in proton radiotherapy. Additionally, the production cross sections are very sensitive to reaction modeling mechanism and associated parameters and measured data can be used to validate existing evaluated nuclear databases, or to improve nuclear reaction models in the energy range up to 200 MeV, especially in the range above 20 MeV where existing data are scarce.

In the context of proton therapy, the Positron Emission Tomography (PET) range verification technique relies on the comparison of the measured and estimated activity

distributions of β^+ emitters produced by the proton beam in the patient, hence relying on the knowledge of the reaction cross sections involved. In order to improve the accuracy with which they are known, we have measured the reaction cross sections that result in ^{11}C ($t_{1/2}=20.36$ min), ^{13}N ($t_{1/2}=9.97$ min) and ^{15}O ($t_{1/2}=122$ s) in the main elements of the human body (C, N and O) in the full energy range of interest, i.e., up to clinical energies of 200 MeV.

The main advantage of proton therapy over conventional radiation therapy is that it offers a better conformation of the dose in the tumor volume while reducing the dose delivered to the surrounding healthy tissues [1]. This is possible due to the particular way in which ions interact with the matter, showing a finite penetration depth (range) at the end of which the dose deposition is maximum, resulting in what is known as the Bragg peak. The mentioned sparing of the healthy tissue makes proton therapy highly recommended for tumors close to organs at risk and in pediatric cases, as it reduces the long-term effects of the irradiation such as the appearance of secondary tumours [2].

In order to fully exploit the benefits of proton therapy it is necessary to make sure that the dose is delivered according to the treatment planning. This can be accomplished with a PET scanner by acquiring the activity map of the patient corresponding to the β^+ decay of the radioactive nuclei produced along the trace of the proton beam [3–5]. The PET image can be acquired during the treatment (in-beam) or after the treatment (offline), detecting the decay of short- or long-lived β^+ emitters, respectively.

The PET measured activity profiles need then to be compared with a calculation, usually made by Monte Carlo simulations, in order to assess if the delivery of the fields has been made according to the treatment planning. Leaving aside biological wash-out effects [6], the accuracy of the simulated activity distribution depends on the underlying cross sections to produce the β^+ emitters of interest. In the case of delayed (offline or in-room) PET verification, the isotopes of interest are ^{11}C , ^{13}N and ^{15}O . At the clinical energies, the main reactions producing these isotopes are $^{12}\text{C}(p,x)^{11}\text{C}$, $^{14}\text{N}(p,x)^{13}\text{N}$ and $^{16}\text{O}(p,x)^{15}\text{O}$. Other reactions of interest are $^{16}\text{O}(p,x)^{11}\text{C}$, $^{16}\text{O}(p,x)^{13}\text{N}$, $^{14}\text{N}(p,x)^{11}\text{C}$ and $^{14}\text{N}(p,\gamma)^{15}\text{O}$ (see Table I for details).

A limitation is that experimental data corresponding to these reactions is quite limited, especially for energies above 20 MeV, and shows sizable discrepancies [7, 8]. Therefore, new measurements and subsequent data evaluations are required in the full energy range of interest in order to reduce uncertainties in the estimation of the activity depth profiles, enabling then the aimed submillimetric beam range verification [9, 10]. Such data needs were highlighted at the IAEA Technical meeting held in Vienna on December 2018 [11].

In this work, the cross sections for the production of the mentioned longer lived β^+ emitters have been measured in the full proton energy range of interest. The range

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TABLE I. Positron emitters considered in this work, listing their half-lives ($t_{1/2}$), branching ratios [13], and the target and specific material foil in which they are produced: polyethylene (PE), silicon dioxide (SiO_2), aluminium nitride (AlN) or copper (Cu).

β^+ emitter	$t_{1/2}$	β^+ emission branching ratio	Target (foil material)
^{11}C	20.36 min	99.9%	C (PE)
^{13}N	9.97 min	99.8%	C (PE), N (AlN), O (SiO_2)
^{15}O	122 s	92.9%	C (PE), N (AlN), O (SiO_2)
^{63}Zn	38.33 min	92.8%	Cu

between 30 and 200 MeV was covered in a single experiment at the West German Proton Therapy Center (WPE) combining the multi-foil activation technique with PET imaging [12]. The lower energy range was assessed at the Heidelberg Ion-Beam Therapy Center (HIT) using conventional scintillation detectors to measure the activation of single foils.

In this paper the experiments are described and the results are presented and compared with the previous data available in the literature. Furthermore, the point-wise cross sections have been fitted to an analytical curve that can be used for PET range verification calculations and could serve as starting point in forthcoming evaluations.

II. IRRADIATION FACILITIES AND DETECTION SYSTEMS

A. West German Proton Therapy Center

1. Beam delivery system at WPE

The West German Proton Therapy Centre (WPE) is a cyclotron-based proton therapy center with four treatments rooms (IBA ProteusPlus) located in Essen, Germany. The isochronous cyclotron accelerates protons to an energy of 228 MeV, which can then be degraded to a minimum of 100 MeV with a system consisting in a wheel-mounted wedge followed by analyzing magnets and energy slits [14]. The experiments were performed in the fixed beam treatment room, where the pencil beam scanning (PBS) mode was chosen in order to deliver the quasi mono-energetic proton beams with less secondary particles compared to the passive beam scanning techniques. The beam is deflected by dipole magnets in order to select the desired lateral position on a time scale in the order of milliseconds. A first ionization chamber is placed at the nozzle entrance checking the centering of the beam, then two monitor chambers are placed close to the nozzle exit to check the spot position and width. Table II summarizes the nominal beam size (sigma for both x- and y-axis) at the isocenter for each energy. The lateral spreading of the beam as it traverses the degraders and target foils has been checked using radiochromic foils.

The delivery beam system allows to fix the total num-

ber of monitor units (MU) for each irradiation (5000 MU). The number of protons per monitor unit (MU) was characterized with a Faraday Cup (FC), whose description can be found in [15, 16], and verified periodically during the quality assurance (QA) procedures in the proton therapy center. Table II summarizes the nominal beam size and accumulated charge for each irradiation.

2. PET/CT scanner at WPE

The induced activity of all the irradiated foils (see Section III A) was determined through a single measurement with a Siemens Healthineers - Biograph Vision PET/CT scanner from the Nuclear Medicine department of the University Hospital Essen, located at the same building than the WPE. This scanner is made of $3.2 \times 3.2 \times 20 \text{ mm}^3$ LSO crystals and silicon photomultiplier (SiPMs) providing a time resolution of just 214 ps, allowing for time-of-flight position reconstruction that improves both contrast and signal-to-noise ratio [17].

The PET/CT scanner is located close to the treatment rooms, which allowed to transport the irradiated foils to the PET scanner rapidly enough to measure the decay of ^{15}O , with a half-life of just 122 s.

The reliability of the PET scanner was assessed with two different experiments:

- A calibrated 1.70(3) kBq ^{22}Na source was placed at 5 different positions and PET images were taken in intervals of 1 and 15 minutes. The short exposures would result in values in agreement within 3.3%, a reproducibility that improved down to 0.7% for the 15 minutes exposures.
- Two $150 \times 30 \times 1 \text{ mm}^3$ graphite sheets were activated with at 150 MeV proton beam to produce an homogeneous (within 1.5% [18]) ^{11}C source. Then, a dynamic PET measurement for 30 minutes of 1 minute interval was acquired selecting as VOIs (Volumes of Interest) the same used later for the experiment (see Section III A). The analysis of the 36 VOIs shows that the activity variations between VOIs is constant within 4.4%, yielding a value of 3.04(13) kBq.

The results from a) show that the PET spatial efficiency is essentially constant. The deviations observed in the activity measurements are merely statistical, due to the limited acquisition times and/or low activities. The outcome from b) confirms the mentioned homogeneity of the PET spacial efficiency, and the 4.4% variations observed are consistent with the contributions from the statistics of 1 minute acquisitions (3% for a few kBq) and the 1.5% inhomogeneities of the irradiation field used to activate the graphite sheet. Overall, as the activities produced in the irradiations discussed later on are in the order of tens of kBq (see Fig. 4), with some of them of just a few kBq, a conservative value of 3% is taken as uncertainty of spatial efficiency.

TABLE II. Details of the performed irradiations at WPE, listing the initial irradiation energy (E_i), the polyethylene (PE) degrader used in each case, beam characteristics, accumulated charge and production yields of the monitor foils at the end of bombardment in each case.

E_i (MeV)	Initial PE degrader	PE degrader between foils	Beam size		Accumulated charge (nC)	Production yield at EOB of the monitor foils (kBq/nC)			Std. Dev.
			σ_x (mm)	σ_z (mm)		PE	SiO ₂	AlN	
100	40 mm	5 mm after the first three foils and 3 mm after the fourth foil	8.08	7.81	64.9(7)	1.16(4)	1.12(4)	1.16(5)	1.8%
100	-	5 mm (2 mm after the last two foils of AlN & SiO ₂ assemblies)	8.08	7.81	64.9(7)	0.92(4)	0.94(4)	0.96(4)	1.7%
150	-	10 mm	5.53	5.23	87.2(9)	0.92(4)	0.94(4)	0.92(4)	0.88%
190	-	10 mm	4.07	4.02	103.0(10)	0.69(3)	0.67(3)	0.68(3)	1.5%

TABLE III. Details of the performed irradiations at HIT, listing the initial irradiation energy (E_i), the polyethylene (PE) degrader used in each case, the two target materials used in each irradiation, beam characteristics and accumulated charge.

E_i (MeV)	PE degrader (mm)	Target 1	Target 2	Beam size FWHM (mm)	Accumulated charge (nC)
50.60	14	PE	PE	31.3	8.9(3)
50.60	15	SiO ₂	SiO ₂	31.3	19.3(6)
50.60	13	AlN	AlN	31.3	19.3(6)
50.60	16	AlN	Cu	31.3	19.3(6)
50.60	16	PE	Cu	31.3	19.3(6)

B. Heidelberg Ion-Beam Therapy Center

1. Beam delivery system at HIT

The Heidelberg Ion-Beam Therapy Center (HIT) is a synchrotron-based particle therapy center with two horizontal treatment rooms, a gantry and an experimental room. The synchrotron accelerates protons to 255 discrete energies between 48 and 220 MeV. After the extraction [19, 20], the beam is sent to one of the rooms and then passed through a monitoring system (BAMS), which consists of three ionization chambers and two multi-wire proportional chambers in the beam line before the exit window [21]. The experiments were performed in the experimental room. The BAMS monitoring system allows to check beam position, width and current. Table III summarizes the nominal beam size (FWHM) at the isocenter. As described before, the lateral spreading of the beam as it traverses the degraders and target foils was checked using radiochromic foils.

The beam current is measured with an ionization chamber (IC) placed at the nozzle before the exit window. The number of protons I_{beam} is determined from the secondary current I_{sec} measured with the IC as [22]:

$$I_{C_{sec}} = \frac{dE}{dx} \cdot \frac{L_{chamber}}{W_{Ar}} \cdot I_{beam}, \quad (1)$$

where the active length of the BAMS-IC ($L_{chamber}$) is 2 cm, the W-value of Argon is 26 eV and $\frac{dE}{dx}$ is the energy loss according to the Bethe-Bloch-formula in the target material of the IC (mainly Ar). Table III summarized the beam current for each irradiation.

2. $\text{LaBr}_3(\text{Ce})$ detectors used at HIT

The activity induced in the irradiated foils at HIT was measured with a conventional γ -ray detection system consisting in two cylindrical 1.5"x1.5" $\text{LaBr}_3(\text{Ce})$ detectors. These were installed in a room next to the irradiation room, so the time span between the end of the irradiation and the start of the measurement is minimized and allows the measurement of the shortest-lived isotope (^{15}O).

The detection efficiency at a specific target-detector distance was determined with an accuracy of 1.4% from a Geant4 [23] Monte Carlo simulation of the β^+ decay, validated experimentally with calibrated ^{137}Cs and ^{22}Na γ -ray sources. This work led to conclude that the $\text{LaBr}_3(\text{Ce})$ crystal size is actually 1.44"x1.44", 12% smaller in volume than indicated by the manufacturer. Ref. [12] provides a detailed explanation of the validation of the corresponding detection and β^+ conversion efficiencies.

III. METHODOLOGY

A. Multi-foil activation and PET/CT measurement at WPE

The employed technique consists in the irradiation of a set of stacks of thin foils made of different materials which activity is then determined in a single PET measurement for all the foils. A detailed explanation and the validation of this technique is given in Ref. [12].

In this work, three stacks (see Fig. 1) made of 10 thin target foils rich in carbon (polyethylene, PE), oxygen (silicon dioxide, SiO₂) and nitrogen (aluminium nitride, AlN) have been irradiated at four different energies, allowing the measurement of the reaction cross sections of interest from ~ 30 MeV up to 200 MeV in small energy steps.

The target thicknesses were 1 mm for the PE and SiO₂ foils and 0.64 mm in the case of AlN (see Table IV for details). For the irradiations the thin foils were embedded between 5 or 10 mm thick PE slabs (see Fig. 1, bottom) in order to degrade the proton beam energy and to have a beam of the desired energy in each foil. As the minimum extracted energy at WPE is 100 MeV, a PE slab of 40 mm was used in front of each stack to degrade the beam energy below 60 MeV. The optimization of the degraders configuration was assessed with Geant4 simulations, and the thicknesses of the PE slabs used in each irradiation

TABLE IV. Detailed information of the target assemblies.

Material	ρ (g/cm ³)	Composition	Thickness (mm)	Reactions channels*
PE	0.96	C ₂ H ₄	1	$^{12}\text{C}(\text{p},\text{x})^{11}\text{C}$
Quartz	2.2	SiO ₂	1	$^{16}\text{O}(\text{p},\text{x})^{11}\text{C}$, $^{16}\text{O}(\text{p},\text{x})^{13}\text{N}$, $^{16}\text{O}(\text{p},\text{x})^{15}\text{O}$
Aluminum nitride	3.3	AlN	0.64	$^{14}\text{N}(\text{p},\text{x})^{11}\text{C}$, $^{14}\text{N}(\text{p},\text{x})^{13}\text{N}$, $^{14}\text{N}(\text{p},\gamma)^{15}\text{O}$
Copper	8.92	Cu	1	$^{nat}\text{Cu}(\text{p},\text{x})^{63}\text{Zn}$ (monitor)

*In bold the main reactions for the production of ^{11}C , ^{13}N and ^{15}O

**Copper target placed always behind SiO₂ assembly

are shown in Table II.

As an illustrative example, the energy probability distributions inside each PE target for each initial beam energy are shown in Fig. 2. The proton energy in each foil is then defined as the mean value of the corresponding distribution, with an asymmetric energy spread calculated from the mean value to the energy for which the probability function reaches half of the maximum.

The PBS delivery mode allows the irradiation of three stacks with the same initial beam energy without entering the treatment room. In this way, the time span between the irradiation of the different targets was practically negligible, and the radioactive decay between the irradiations and the PET measurement was minimized. The alignment and targets positioning at the isocenter was carried out with the help of the laser positioning system. In front of each assembly there was a 1 mm thick PE foil used for monitoring purposes: the activity induced in the monitor foil of the three stacks must be proportional to the beam charge accumulated by each stack. Then, for validation purposes, a 2 mm ^{nat}Cu target was placed behind the SiO₂ stack in order to measure the production yield of the IAEA standard reaction $^{nat}\text{Cu}(\text{p},\text{x})^{63}\text{Zn}$ ($t_{1/2}=38.33$ min) [24] (see Fig. 1, bottom).

The details of each irradiation (initial energy, degraders and accumulated charge) are summarized in Table II. After each irradiation with a given beam energy, the activity from all the irradiated foils was measured simultaneously using the PET scanner: a total of 36 foils from the PE, SiO₂, AlN, the monitor foils and Cu foils were inserted into a 16x32 cm² PE matrix, which serves both as positioning system inside the PET scanner and conversor of the positrons into two 511 keV photons. This is necessary because the range of the positrons emitted by ^{11}C , ^{13}N and ^{15}O is of a few meters in air, and only 5, 6 and 9 mm, respectively, in PE. Fig. 3 shows: (top) the disposition of the irradiated foils in two layers within the PE matrix, (bottom) the complete PE matrix inside the PET scanner.

The PET/CT scanner was operated in dynamic mode to obtain the activity in each foil as a function of time, recording images of 1 minute during 4 hours, time enough to reach the background level. Two different reconstruction protocols (iterative and TrueX, both with Time of Flight algorithm) were used, providing compatible results within always less than 1%. The images obtained include the corresponding attenuation corrections within the PE matrix according to the computerized tomographies (CT) images taken with the PET/CT scanner. The image pro-

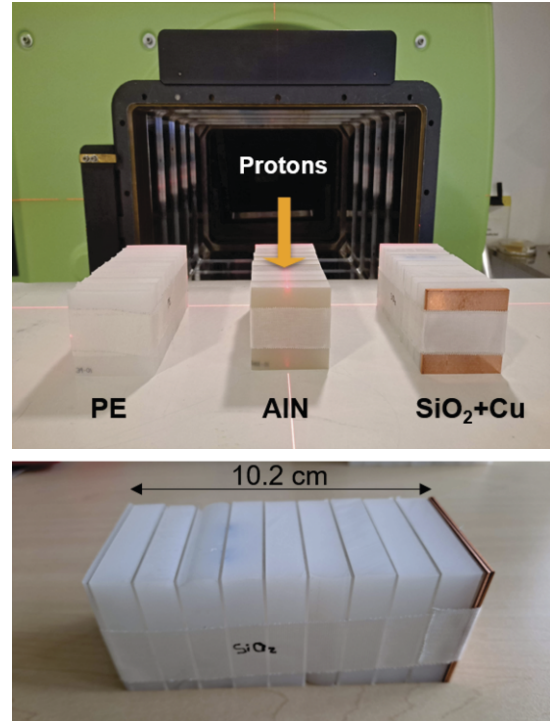


FIG. 1. (Color online) Top: Experimental setup during the irradiations at WPE. Bottom: The SiO₂ stack made of 10 thin foils of SiO₂ between thicker PE degraders. The first and last foils are the PE and copper monitor foils, respectively.

cessing software PMOD [25] allows quantitative analysis of the images, defining the corresponding Volumes-of-Interests (VOIs) for each foil and obtaining the number of counts per second within each VOI as a function of time. The calibration from number of counts per second to activity units was achieved by measuring the calibrated 1.70(3) kBq ^{22}Na source inside the PE matrix.

B. Low-energy activation measurements using conventional scintillator detectors at HIT

The minimum initial beam energy available at WPE is 100 MeV, which means that the energy spread is too large when one degrades the beam down to ~ 30 MeV. Therefore the measurements at such low energies were performed at the HIT facility, which can deliver proton

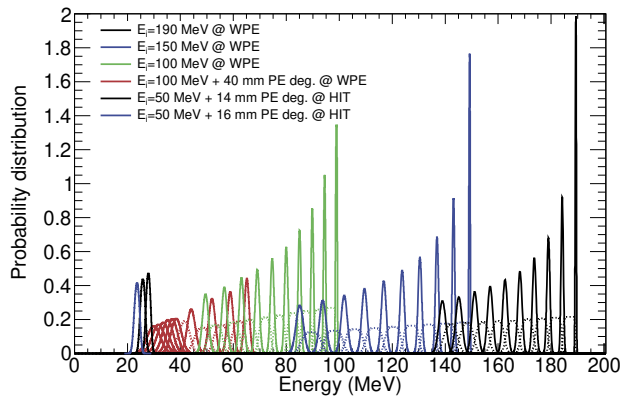


FIG. 2. (Color online) Proton energy distribution inside PE foils at different initial beam energies for the WPE and HIT irradiations. Dotted lines: Proton energy distributions inside PE degraders placed between target foils.

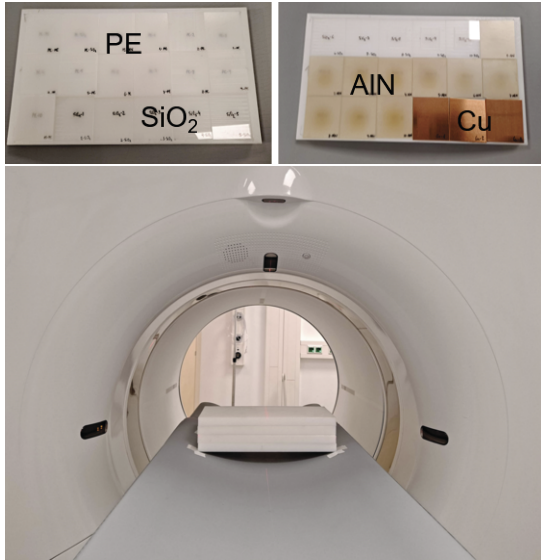


FIG. 3. (Color online) Up: Positioning of the irradiated foils between the polyethylene thick layers. Bottom: PET measurement of the activity of the irradiated foils embedded in the PE matrix.

beams with a minimum of 50 MeV. Unfortunately, HIT does not host a PET/CT scanner and hence a different approach had to be followed for measuring the induced β^+ activities.

As in WPE, the proton energy was degraded using PE layers with different thicknesses to achieve the desired proton energy distribution inside each foil. The above-mentioned Fig. 2 displays as well the energy probability distribution in the PE foils measured at HIT, illustrating the improved energy resolution achieved at HIT with respect to WPE in the low proton energy range. The production yields in carbon and nitrogen are obtained at three energy points, and two in the case of oxygen. Fol-

lowing the same strategy as at WPE, a copper target was irradiated together with a PE target and a AlN target in order to validate the analysis and corrections applied using the IAEA standard reaction $^{nat}\text{Cu}(p,x)^{63}\text{Zn}$ [24] below 20 MeV. Table III summarizes the details of each irradiation: initial energy, targets, degraders in front of each target and accumulated charge.

After each irradiation, the activity in the corresponding foil was studied with the LaBr_3 γ -ray detectors discussed in Section II B 2. For this, each target was sandwiched between a pair of 1 mm thick lead foils, acting as positron converters, and placed at 150 mm from the front face of the detector. The detectors were connected to a CAEN V1750 digitizer that provided the signals' integral with the corresponding timestamp, allowing then to study the decay curve from the positron emitters of interest.

C. Analysis of the activity curves

As the outcome of the two experiments is the induced activity curve in each target as a function of time, the data analysis for both experiments is quite similar: it consists in the determination of the induced activity at the end of the irradiation and, from that value, the calculation of the production yields and the corresponding cross sections for the corresponding energy and reaction.

The differential equation describing the process of radioisotope formation (f_{prod}) and decay during the irradiation ($-\lambda N$) assuming the thin target approximation is:

$$\frac{dN}{dt} = f_{\text{prod}} - \lambda N = n_i \sigma I - \lambda N, \quad (2)$$

where λ is the corresponding decay constant, n_i is the surface density of foil i in units of atoms per unit area, I is the proton beam current and σ is the cross section, considered constant for each foil i .

The limits of this approximation have been studied at the lower energies, i.e., where the spread of the energy distributions reaches its maximum value and there is a sharp increase in the cross sections. In the experiment described in Sec. III A, the comparison of the cross sections at the entrance/exit of the last foil with the cross section in the full target is always less than 10%, then allowing the thin target approximation in this limit case.

Solving this equation for $N(0) = 0$, the number of produced nuclei at the end of bombardment (EOB) is:

$$N = n_i \sigma I \cdot \frac{1 - e^{-\lambda t_{\text{irr}}}}{\lambda}. \quad (3)$$

If $\lambda t \ll 1$ ($t_{\text{irr}} \ll t_{1/2}$), then equation 3 becomes:

$$N_{\lambda t \ll 1} = n_i \sigma I t_{\text{irr}}, \quad (4)$$

thus the fraction of nuclei that decayed during the irradiation can be defined as:

$$c_{\text{decay}} = \frac{N_{\lambda t \ll 1}}{N} = \frac{\lambda t_{\text{irr}}}{1 - e^{-\lambda t_{\text{irr}}}}. \quad (5)$$

Taking into account that the radioisotope activity is $A = \lambda N$, the production yield Y (activity per unit charge, C_{total}) in each foil is calculated as:

$$Y = \frac{A^{EOB}}{C_{total}} \cdot c_{decay}. \quad (6)$$

From the previous expressions, the production cross section at the energy E_i of the beam traversing foil i , from the corresponding yield Y_i , can be expressed as:

$$\sigma(E_i) = \frac{Y_i(E_i)}{\lambda \cdot n_i} = \frac{A_i^{EOB}}{C_{total} \cdot \lambda \cdot n_i} \cdot c_{decay}. \quad (7)$$

In both experiments, the measured quantity was the counting rate as a function of time, which has been converted into an activity decay curve $A(t)$ by dividing it by the detection efficiency (of the PET or the LaBr₃ in each case) and by the intensity associated to the decay (see Table I). The corresponding activity value at the end of the bombardment A^{EOB} , was determined by fitting each decay curve to the expected exponential decay:

$$A(t) = C + A_{11C}^{EOB} e^{-\lambda_{11C} t} + A_{13N}^{EOB} e^{-\lambda_{13N} t} + A_{15O}^{EOB} e^{-\lambda_{15O} t}. \quad (8)$$

Given the large count statistics acquired and the sufficient length of the measurements that allow reaching the background level, the uncertainty in the fit is in most cases negligible, i.e. less than 1% (fit and uncertainty given by ROOT [26], see Fig. 4). The total uncertainty in the yield is calculated as detailed in Sec. IV.

D. Cross-checking with monitor foils

As mentioned in section III A, the first foil in each stack of the multi-foil experiment at WPE for each irradiation was a PE foil used for monitoring and validation purposes. These monitor foils were measured simultaneously with the rest of the foils with the PET/CT scanner.

The decay curves of the three monitor foils for 150 MeV irradiation are displayed in Fig. 5. The corresponding production yield of ^{11}C in the total of the twelve monitor foils used are summarized in Table II and are in a very good agreement (less than 2% in the four irradiations), providing evidence of the reliability of the relative beam current measurement.

E. Validation with the IAEA monitor reaction

In both experiments, a ^{nat}Cu target was irradiated together with at least one of the target foils. This allows to measure the integral production yield below 20 MeV and compare it with the value expected from the IAEA evaluation [24].

At WPE, a 3 mm of ^{nat}Cu target was irradiated together with the SiO_2 assembly during the first irradiation

listed in Table II. At HIT, two Cu targets of 1 mm were irradiated behind a PE and a AlN targets, respectively. The activities were measured as function of time with the PET scanner and LaBr₃ detectors, respectively, and the corresponding activity curves were fitted to the sum of three exponentials, each one corresponding to the production of ^{63}Zn ($t_{1/2}=38.47$ min), ^{62}Cu ($t_{1/2}=9.673$ min) and ^{62}Zn ($t_{1/2}=9.186$ h). As an example, the activity curve and the corresponding fit of the WPE experiment are shown in Fig. 6. The experimental production yields were then calculated from Eq. 6.

Regarding the expected ^{63}Zn production yields, as the energy of the protons hitting the copper target is not mono-energetic, they were calculated as the convolution of the IAEA production yield as function of the proton energy [24] with the proton energy distribution in the Cu targets given by Geant4.

The expected and measured thick target physical yields for each Cu irradiation, at both WPE and HIT, are given in Table V. In both cases, the good agreement (in the order of 3-6%) between the expected and the measured values are compatible with the $\sim 4\%$ accuracy of the experimental value (see Sec. IV).

Considering that the $^{nat}\text{Cu}(p,x)^{63}\text{Zn}$ cross section is an IAEA monitor cross section featuring low uncertainties, these results serve as validation of the measuring technique, the analysis strategy and corrections applied for both experiments; it validates the beam current measurements as well as the detection efficiency calculation.

TABLE V. Comparison between the IAEA and experimental production yields of the $^{nat}\text{Cu}(p,x)^{63}\text{Zn}$ reaction

Experiment	Target with which it is irradiated	Y_{exp} (kBq)	Y_{theo} (kBq)	Ratio
WPE	SiO_2	2.72(11)	2.62(11)	1.04
HIT	PE	2.28(8)	2.38(10)	1.04
HIT	AlN	2.16(7)	2.28(9)	1.06

IV. RESULTS AND DISCUSSION

The cross sections for all the measured reactions have been calculated from the production yields using Eq. 7. The obtained point-wise cross section are presented in Tables VI, VII and VIII in V A, where the quoted uncertainties have contributions from the fit, the spatial dependence of the PET scanner efficiency (3%, see Section II A 2), the activity of the ^{22}Na calibration source (1.5%), the uncertainty in the foil thicknesses (1%) and the accuracy of the measurement in the beam current (1% in the measurements at WPE [15] [16] and less than 3% in all cases in the measurements at HIT [20]). The accuracy of the measurements ranges between $\sim 2.3\%$ -8.9%.

As the data presented herein have been obtained using a multi-foil activation technique, the very good agreement between the first and last point of two consecutive irradiation energies serves also as cross-check of the results.

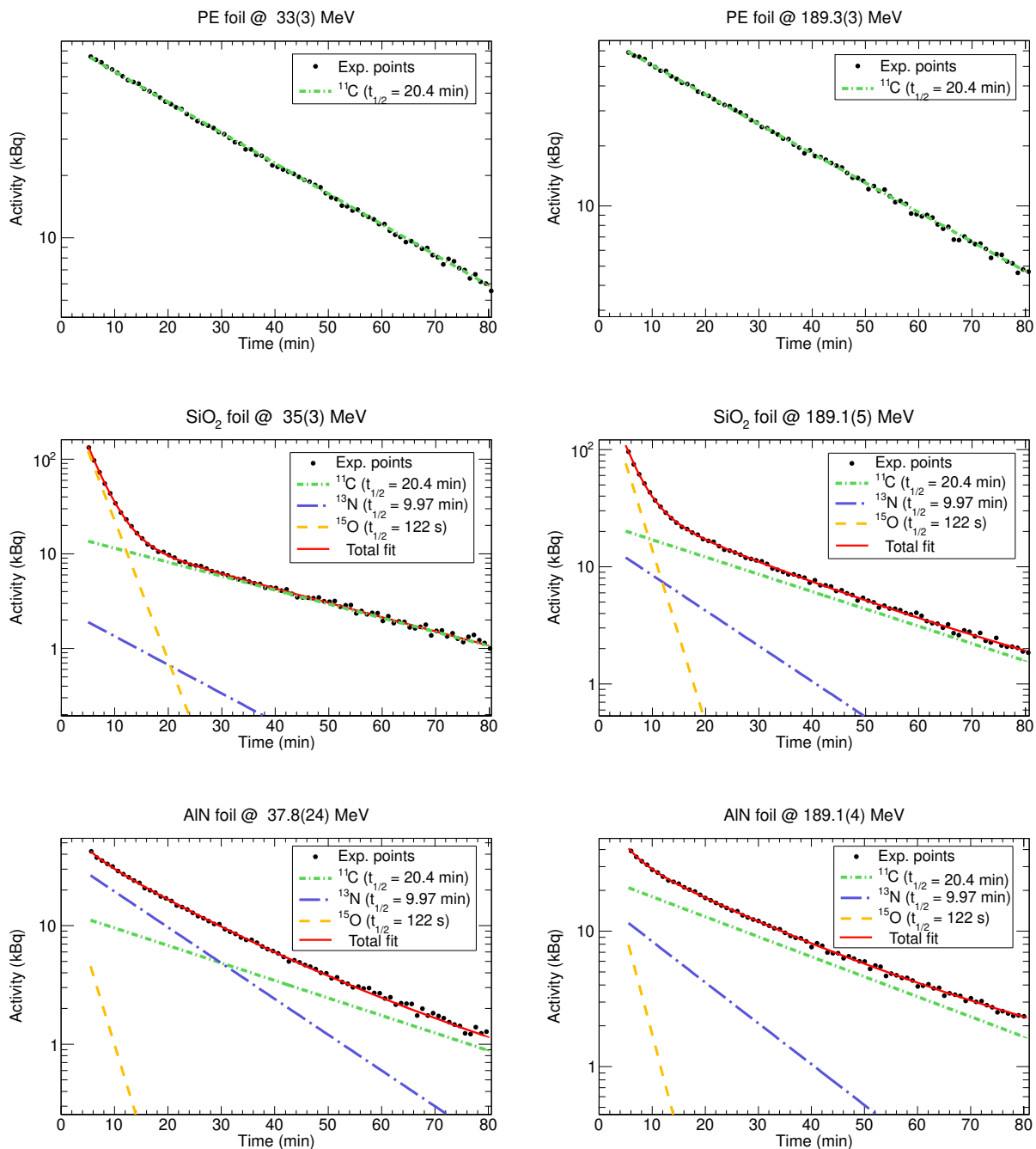


FIG. 4. (Color online) Induced activity distribution in full VOI of PE (top), SiO_2 (middle) and AlN (bottom) foils at low ($E \sim 35$ MeV) and high ($E \sim 190$ MeV) energy including exponential fits. For the sake of clearness, the decay curves are displayed only up to 80 minutes, although they are fitted up to 300 minutes.

The measured cross sections are displayed as well in the top panel of Fig. 7, while the bottom panel shows the cross sections weighted by the percentage of the number of atoms of C (12%), N (1.1%) and O (24%), relative to the total number of atoms in the human body, in order to illustrate their importance for beam range verification. Fig. 7 also displays the parametrization of the cross sections

to the analytical curve:

$$f(x) = \frac{\sum_{i=0}^4 a_i \cdot x^i}{x^5 + \sum_{j=0}^4 b_j \cdot x^j}, \quad (9)$$

which can be used for PET range verification calculations and could serve as starting point in forthcoming evaluations. The parameters a_i and b_j of the fit for each reaction

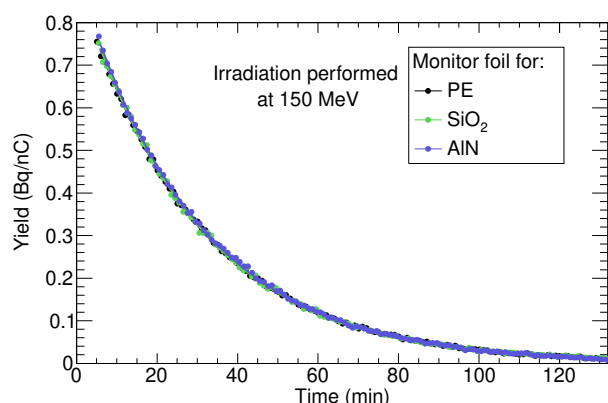


FIG. 5. (Color online) Induced activity distribution per total charge of the PE monitor foils at $E_i = 150$ MeV.

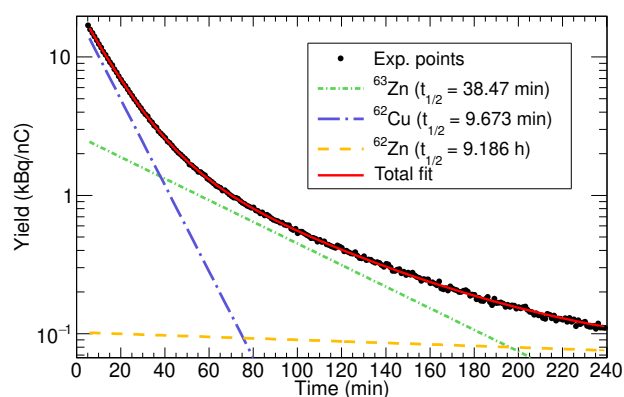


FIG. 6. (Color online) Induced activity distribution per total charge in the ^{nat}Cu target at WPE.

are attached in Appendix V B.

In the following, the results of this work for the production of ^{11}C , ^{13}N and ^{15}O on C, N and O are presented and discussed separately, comparing in each case our results with those available in the literature, both the experimental and evaluated data. Note that the ICRU63 library [27] is the extension to 250 MeV of ENDF/B-VII.0 (LA-150) library for several isotopes of particular interest in proton therapy (as the ones studied in this work), whereas TENDL-2019 includes ENDF/B-VII.0 data whenever they are available, which is also the situation in the present study. Therefore in the following we will refer to all of them as the ENDF/B-VII.0 library.

A. Production of ^{11}C in C, O and N

There are quite some data sets in the literature for the $^{12}\text{C}(\text{p,pn})^{11}\text{C}$ reaction, as it dominates the production of ^{11}C in the human body at the typical energies of proton therapy, but none of them until now covers the full energy

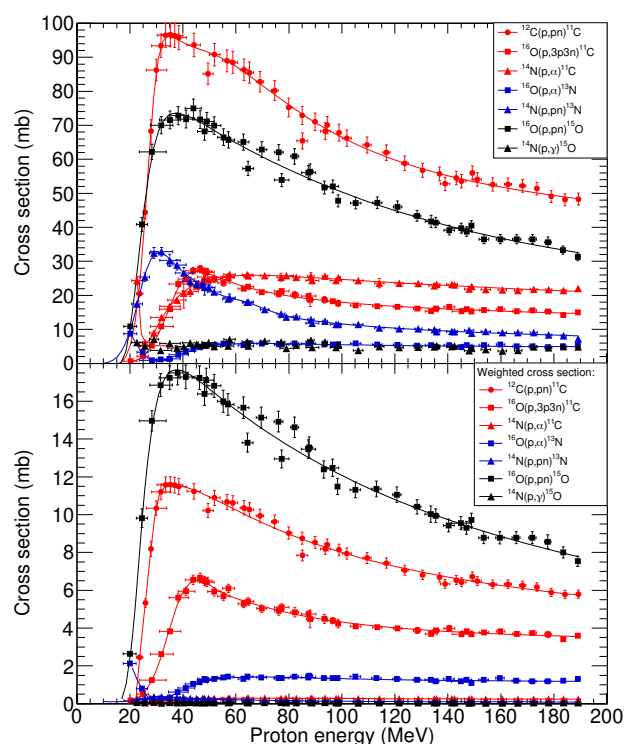


FIG. 7. (Color online) Top: Proton-induced reaction cross section to produce ^{11}C , ^{13}N and ^{15}O in C, N and O. Bottom: Proton-induced reaction cross section to produce ^{11}C , ^{13}N and ^{15}O weighted by the C, N and O composition in human body.

range of interest. From the data displayed in Fig. 8 one can observe two main groups: one including the data of Hintz et al. (1952) [28], Aamodt et al. (1952) [29], Bäumer et al. (2019) [16] and Bäcker et al. (2019) [30], and another one with Gauvin et al. (1962) [31], Matsushita et al. (2016) [32] or Horst et al. (2019) [33]; the latter being about 15 to 25% lower than the first, even though there are recent works in both groups. It is noticeable that the two most recent data sets, reported by Bäumer et al. [16] and Horst et al. [33], show differences, partially resolved in [34], beyond 15% at about 100 MeV. Furthermore, all measurements indicate an underestimation of this cross section in the ENDF/B-VII.0 library.

This situation illustrates the need of new data with enhance accuracy covering the full range of interest, as done in this work.

The cross section reported herein is in very good agreement with Hintz et al. (1952) [28] and Aamodt et al. (1952) [29], and also with the recent work by Bäumer et al. (2019) [16] and Bäcker et al. (2019) [30], i.e. with the first group of measurement mentioned in the previous paragraph.

The $^{16}\text{O}(\text{p,x})^{11}\text{C}$ production channel has been less investigated in previous studies, but it is also of interest in PET range verification, as about the 65% of the human body is composed by oxygen. Indeed, Fig. 7 (bottom)

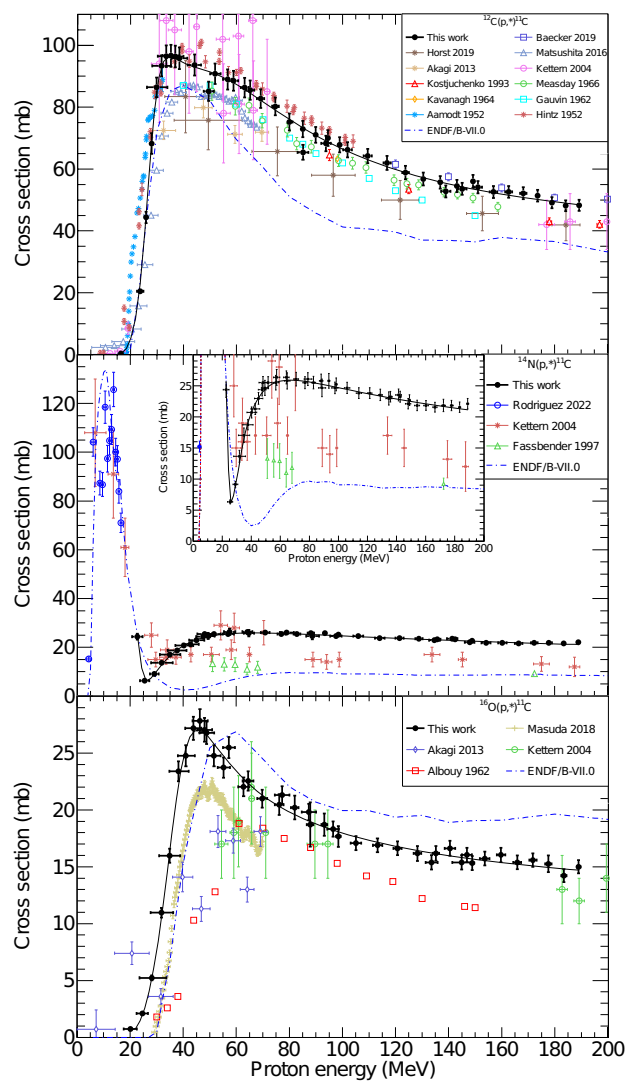


FIG. 8. (Color online) Cross sections for the production of ^{11}C on C, N and O. Note that, although the original evaluated data correspond to the ENDF/B-VII.0 library, in order to cover our full energy range we have included also their ICRU63 extrapolation, which originates the change of style above 150 MeV.

shows that its contribution to the production of ^{11}C in the human body is about half of the production in carbon, hence not negligible, thus supporting the importance of measuring the production cross sections, not only in carbon, but in the other main elements of the human body.

As illustrated in Fig. 8, the previous data from Albouy et al. (1962) [35], Kettern et al. (2004) [36], Akagi et al. (2013) [37] and Masuda et al. (2018) [38] show significant discrepancies, reaching differences up to 40% at the low energy region (~ 50 MeV) and even larger discrepancies right above the reaction threshold. The new data presented herein shows a reasonable agreement with the only data available around 200 MeV (Kettern et al.), which con-

firms a $\sim 25\%$ overestimation of the ENDF/B-VII.0 data. In the 100 to 150 MeV region our data are about 20% larger than the only data available (Albouy et al.). Last, at lower energies, our cross section is in general higher than all others, specially right above the threshold at ~ 30 MeV and up to ~ 60 MeV, which is the region closest to the Bragg peak and hence of most importance.

Nitrogen is the fourth most abundant element in the body after oxygen, carbon, and hydrogen. However, although the production of ^{11}C in N is not as significant as the production in C or O, at high energies, the $^{14}\text{N}(p,x)^{11}\text{C}$ reaction features a threshold of only 3.1 MeV, which makes it the dominant reaction for ^{11}C production below the reaction thresholds around 20 MeV on C and O, hence at the very end of the beam range. While the low energy region has been studied in a previous work of ours [12] and there is actually an IAEA evaluation below 30 MeV, the measurement at higher energies reported in this work is important for having a complete and accurate data base.

As it can be seen in Fig. 8, there are only two data sets for this reaction between 20 and 200 MeV, being our data well above those of Fassbender et al. (1997) [39] and, compared to the recent data of Kettern et al. (2004) [36], not far below ~ 70 MeV but as much as a factor two larger beyond 100 MeV.

B. Production of ^{13}N in O and N

The production of ^{13}N in the human body is dominated by its production in nitrogen (see Fig. 7, bottom). However, the $^{14}\text{N}(p,pn)^{13}\text{N}$ reaction channel has not been investigated much, being there only three data sets: Kettern et al. (2004) [36], Sajjad et al. (1986) [40] and Valentin et al. (1965) [41]. As shown in Fig. 9, our data is in good agreement with Valentin et al. [41], while the quite recent data of Kettern et al. seems to have an issue because it falls too rapidly to negligible values at a few tens of MeV. Below 30 MeV, the two previous data sets of Sajjad et al. and Valentin et al. show considerable difference in the estimation of the reaction threshold, and thus would show a difference in the activation profiles for PET range verification. The data presented herein at low energy constitute a new data set and agree with Valentin et al., solving the discrepancy.

Regarding the production in oxygen via $^{16}\text{O}(p,x)^{13}\text{N}$ reactions, there is actually an IAEA evaluation below 30 MeV, against which the technique presented herein was validated in a previous work [12]. However, because most previous studies were motivated by the production of radioisotopes for PET diagnostic, data above 30 MeV are not so prolific. Comparing our results in Fig. 9 with the four data sets available (Foley et al. (1962) [42] (only one point), Albouy et al. (1962) [35], Valentin et al. (1963) [43] and Kettern et al. (2004) [36]): in the region between 20 and 40 MeV, the data presented herein are considerable lower than Albouy et al.; from 40 to 200 MeV, our data are about a $\sim 25\text{-}30\%$ higher than Albouy et al., Kettern

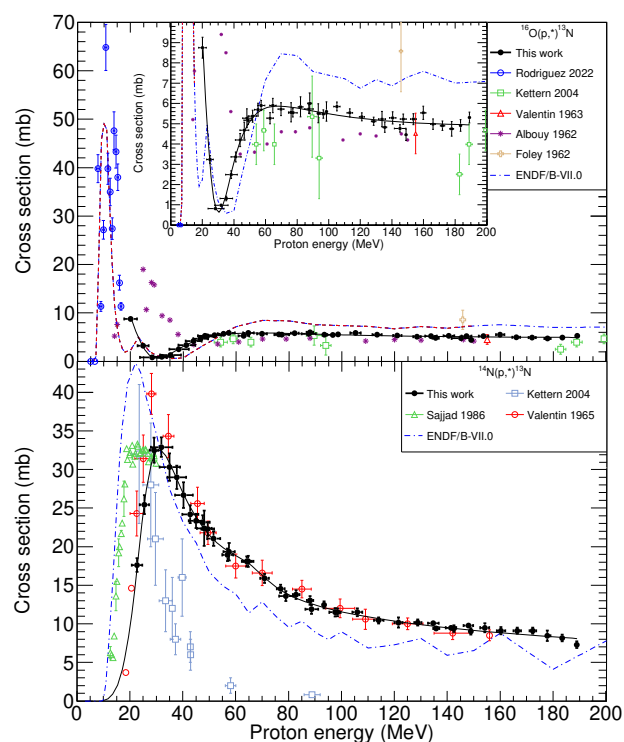


FIG. 9. (Color online) Cross sections for the production of ^{13}N on N and O. Note that, although the original evaluated data correspond to the ENDF/B-VII.0 library, in order to cover our full energy range we have included also their ICRU63 extrapolation, which originates the change of style above 150 MeV.

et al. and Valentín et al, but almost a $\sim 50\%$ lower than Foley et al. The low production yields constitute a reason why the cross-section points range around the fitting curve.

C. Production of ^{15}O in O and N

In delayed PET range verification, the biological washout of the generated isotopes plays an important role in the feasibility of the technique. Isotopes with shorter half-lives are recommended to avoid this effect. For instance, reducing the time span between irradiation and measurement to detect ^{15}O , which is produced mainly by the $^{16}\text{O}(p, x)^{15}\text{O}$ reaction (see Fig. 7, bottom), would reduce wash-out effects. The increase in the interest in using this isotope brings about new measurements, such as in Horst et al. (2019) [33], Masuda et al. (2018) [38] and Akagi et al. (2013) [37], to complete the previous ones from Sajjad et al. (1985) [44], Valentín et al. (1965) [41] and Albouy et al. (1962) [35]. It seems that in this case, see Fig. 10, our data and most previous measurements are in good agreement, especially below ~ 80 MeV, while above that energy our data are slightly lower than all others. It is

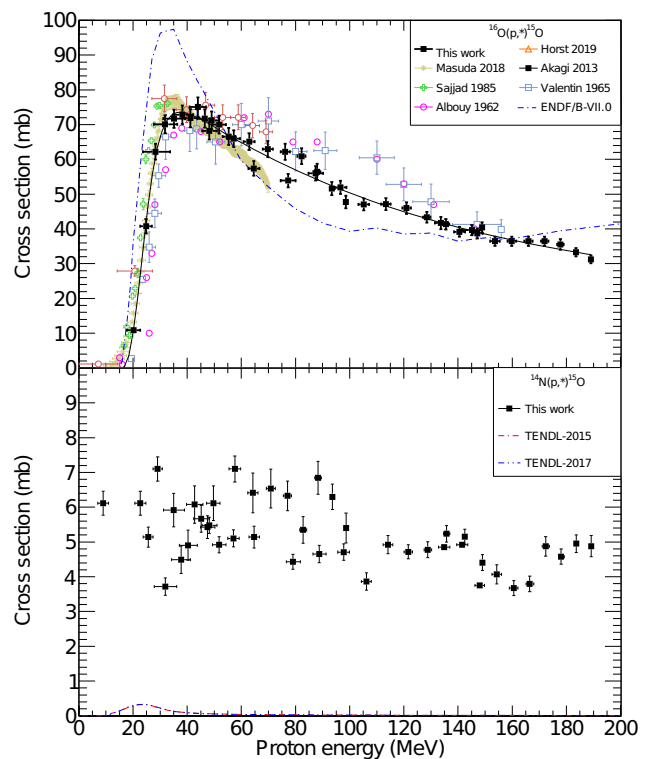


FIG. 10. (Color online) Cross sections for the production of ^{15}O on N and O. Note that, although the original evaluated data correspond to the ENDF/B-VII.0 library, in order to cover our full energy range we have included also their ICRU63 extrapolation, which originates the change of style above 150 MeV.

noticeable that the shape of the ENDF/B-VII.0 library is quite far from all experimental data, a difference that has considerable effects in the activation profiles estimations necessary for PET range verification.

Last, although its role in this context is very limited, for the sake of completeness, the production of ^{15}O in nitrogen has been obtained as well. In this particular case, there is no previous measurement in the literature, hence the importance of the new data set obtained in this work. The evaluated data are not available in ENDF/B-VII.0 library (and therefore neither they are in ICRU63 and TENDL-2019). The only data available are in older TENDL libraries, calculated with TALYS code, which offer in general less confidence.

V. SUMMARY AND CONCLUSIONS

In the last few years, there is an increased interest in the proton therapy community to improve the dose delivery by means of PET range verification [45, 46]. The reliable implementation of this technique requires accurate production cross section data, specifically for the long-lived positron emitters ^{11}C , ^{13}N and ^{15}O in the

energy range up to 200 MeV. In this context, new measurements with minimized uncertainties should lead to higher accuracy in the cross sections used in the Monte Carlo calculations which determine the β^+ activity distributions in the body needed for PET range verification [8]. Furthermore, these data are also extremely useful for the development of more precise reaction models and to improve existing evaluated databases.

The experimental method employed in this work combines the common multi-foil activation technique with the use of a PET scanner as detector. This allows measuring the activity of all the foils at the same time in a single PET acquisition, hence minimizing the number of irradiations needed and possible systematic errors in the irradiation and in the subsequent activity determination.

Several actions have been implemented successfully in order to validate the results obtained with this new technique: i) a PE foil has been placed in front of each multi-foil stack (agreement within 1.8% between different irradiation and beam energies), ii) the IAEA monitor reaction cross section of $^{nat}\text{Cu}(\text{p},\text{x})^{63}\text{Zn}$ was measured (3% agreement with the evaluated cross section), iii) the beam energies were chosen such that some data points are measured twice (agreement within uncertainties).

The multi-foil targets could not be employed to study very low (below 30 MeV) energies. Therefore, a set of single irradiations and production yield measurement with conventional detectors, instead of the PET scanner, has been performed, using again the evaluated $^{nat}\text{Cu}(\text{p},\text{x})^{63}\text{Zn}$ production cross section for validation (agreement better than 6%).

The outcome of this work is a total of seven excitation functions (40 energy points/reaction) from 30 up to 200 MeV. The production of ^{11}C , ^{13}N and ^{15}O in the main elements of the human body and up to 200 MeV, via the reactions $^{12}\text{C}(\text{p},\text{x})^{11}\text{C}$, $^{16}\text{O}(\text{p},\text{x})^{15}\text{O}$, $^{16}\text{O}(\text{p},\text{x})^{11}\text{C}$,

$^{16}\text{O}(\text{p},\text{x})^{13}\text{N}$, $^{14}\text{N}(\text{p},\text{x})^{11}\text{C}$, $^{14}\text{N}(\text{p},\gamma)^{15}\text{O}$ and $^{14}\text{N}(\text{p},\text{x})^{13}\text{N}$, have been measured. In addition to the point-wise cross section, the excitation functions have been parameterized so that they can be easily implemented in simulations codes for PET range verification. In this line, work is ongoing for a detailed study of the production rates of the β^+ emitters of interest in tissue-equivalent phantoms using the cross sections from this work and their comparison with the activity distributions measured in realistic clinical scenarios, and preliminary studies illustrate the importance of new data and the subsequent data evaluations in the full energy range of interest in proton therapy for a reliable implementation of PET range verification.

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APPENDIX

A. Production cross sections

TABLE VI. $^{12}\text{C}(\text{p},\text{x})^{11}\text{C}$ experimental cross sections.

Facility	E (MeV)	σ (mb) $^{12}\text{C}(\text{p},\text{x})^{11}\text{C}$
HIT	23.6(12)	20.5(8)
	25.8(11)	44.4(18)
	27.9(10)	68(3)
WPE	30(3)	87(3)
	32(3)	94(3)
	33(3)	97(4)
	35.2(24)	97(4)
	36.9(23)	97(4)
	38.4(23)	96(3)
	44.1(18)	94(3)
	51.9(15)	91(3)
	58.2(13)	89(3)
	68.1(11)	86(3)
	$49.5^{+1.4}_{-1.3}$	86(3)
	56.6(12)	90(3)
	63.1(11)	87(3)
	69.1(10)	83(3)
	74.7(8)	81(3)
	80.0(8)	76(3)
	85.0(7)	73(3)
	89.8(6)	72(3)
	94.4(4)	71(3)
	98.9(4)	68.3(25)
	85.1(17)	66.1(24)
	93.7(15)	68.9(25)
	101.9(14)	66.9(24)
	109.5(12)	64.8(23)
	116.8(11)	62.6(23)
	123.8(10)	59.5(21)
	130.5(8)	57.3(21)
	136.9(7)	56.3(20)
	143.1(5)	55.1(20)
	149.2(3)	56.6(20)
	138.9(15)	53.5(19)
	145.1(14)	54.3(20)
	151.1(13)	54.9(20)
	157.0(12)	53.3(19)
	$162.6^{+1.1}_{-1.2}$	53.4(19)
	168.2(10)	52.9(19)
	173.6(9)	52.1(19)
	179.0(7)	49.8(18)
	184.2(5)	48.8(18)
	189.3(3)	48.9(18)

TABLE VII. $^{16}\text{O}(\text{p},\text{x})^{11}\text{C}$, $^{16}\text{O}(\text{p},\text{x})^{13}\text{N}$ and $^{16}\text{O}(\text{p},\text{x})^{15}\text{O}$ experimental cross sections.

Facility	E (MeV)	σ (mb)		
		$^{16}\text{O}(\text{p},\text{x})^{11}\text{C}$	$^{16}\text{O}(\text{p},\text{x})^{13}\text{N}$	$^{16}\text{O}(\text{p},\text{x})^{15}\text{O}$
HIT	$20.2^{+2.5}_{-2.4}$	0.69(3)	8.7(5)	10.9(5)
	24.8(21)	2.08(11)	3.24(19)	40.8(21)
WPE	28^{+4}_{-3}	5.3(2)	0.82(7)	62.6(23)
	32(3)	11.0(4)	0.98(8)	70(3)
	35(3)	16.1(6)	1.32(11)	72(3)
	$38.2^{+2.5}_{-2.4}$	23.5(9)	2.51(19)	73(3)
	$41.2^{+2.1}_{-2.0}$	24.9(9)	3.39(25)	72(3)
	43.9(19)	27.3(10)	4.2(3)	75(3)
	$46.5^{+1.9}_{-1.8}$	28.0(10)	4.7(4)	72(3)
	$49.0^{+1.8}_{-1.7}$	27.0(10)	5.20(21)	72(3)
	57.2(14)	25.6(9)	5.94(25)	66.2(24)
	64.6(13)	22.7(8)	5.9(3)	57.8(21)
	$48.2^{+1.8}_{-1.7}$	27.2(11)	5.3(5)	68.7(25)
	$51.8^{+1.6}_{-1.5}$	24.9(9)	5.4(3)	70(3)
	55.3(15)	23.8(9)	5.7(3)	67.0(24)
	62.9(13)	22.1(8)	5.3(3)	65.5(24)
	69.7(12)	21.1(8)	5.8(3)	63.4(23)
	$76.2^{+1.1}_{-1.0}$	20.6(9)	5.6(5)	62.6(23)
	82.2(9)	20.3(11)	5.9(5)	61.4(23)
	87.9(8)	18.8(20)	6.0(5)	56.8(24)
	93.3(8)	18.8(10)	5.6(5)	52.0(19)
	98.6(7)	17.8(9)	5.6(5)	48.1(18)
	77.3(21)	21.5(8)	5.60(21)	54.4(20)
	87.4(19)	20.0(7)	5.78(21)	56.6(20)
	96.6(16)	18.5(7)	5.54(20)	55.5(19)
	105.2(14)	17.2(6)	5.91(22)	47.5(17)
	113.3(12)	17.0(6)	5.59(20)	47.6(17)
	121.0(11)	16.8(6)	5.40(20)	46.5(17)
	128.4(9)	16.3(7)	5.17(20)	43.8(16)
	135.5(8)	16.3(6)	4.88(25)	41.8(15)
	142.3(6)	15.5(6)	4.82(19)	40.1(15)
	148.9(5)	15.4(6)	4.5(3)	40.8(15)
	133.7(17)	15.6(6)	5.3(4)	42.3(18)
	140.5(15)	16.8(7)	5.4(3)	39.7(14)
	147.2(15)	16.2(6)	5.3(3)	39.3(14)
	153.6(13)	15.9(6)	5.3(3)	37.0(13)
	159.9(12)	16.2(6)	5.6(3)	37.0(13)
	166.0(11)	15.6(6)	5.0(3)	37.0(13)
	$172.0^{+1.0}_{-0.9}$	15.8(6)	4.97(23)	37.0(13)
	177.8(8)	15.5(7)	4.8(4)	36.0(15)
	183.5(6)	14.4(6)	5.0(3)	33.7(12)
	189.1(5)	15.2(6)	5.4(3)	31.7(11)

TABLE VIII. $^{14}\text{N}(\text{p},\text{x})^{11}\text{C}$, $^{14}\text{N}(\text{p},\text{x})^{13}\text{N}$ and $^{14}\text{N}(\text{p},\text{x})^{15}\text{O}$ experimental cross sections.

Facility	E (MeV)	σ (mb)		
		$^{14}\text{N}(\text{p},\text{x})^{11}\text{C}$	$^{14}\text{N}(\text{p},\text{x})^{13}\text{N}$	$^{14}\text{N}(\text{p},\gamma)^{15}\text{O}$
HIT	22.7(20)	24.3(12)	17.5(9)	6.1(3)
	25.6(18)	6.1(3)	25.4(13)	5.1(3)
	29.1(17)	9.0(5)	32.5(17)	7.1(4)
WPE	32(3)	13.6(9)	33.0(21)	3.7(3)
	35(3)	17.0(12)	30.5(14)	6.0(5)
	37.8(24)	18.8(8)	29.2(11)	4.5(3)
	$40.3^{+2.3}_{-2.2}$	20.8(9)	26.8(11)	4.9(3)
	$42.7^{+2.1}_{-2.0}$	21.3(9)	24.3(10)	6.1(5)
	45.1(19)	22.9(9)	23.5(14)	5.7(3)
	47.5(17)	24.3(10)	23.3(9)	5.5(3)
	49.6(16)	24.7(12)	22.4(11)	6.2(3)
	57.6(14)	26.4(13)	19.4(10)	7.2(3)
	64.7(12)	25.9(13)	18.1(9)	5.2(3)
	$48.2^{+1.8}_{-1.7}$	25.5(10)	22.5(8)	5.5(5)
	51.8(16)	25.5(10)	21.2(8)	5.0(4)
	57.0(14)	25.5(12)	19.0(7)	5.1(3)
	64.2(12)	26.3(11)	18.1(8)	6.4(5)
	70.9(11)	26.0(11)	16.0(7)	6.5(5)
	77.0(9)	25.5(10)	14.6(5)	6.4(4)
	82.8(8)	25.5(10)	13.8(6)	5.4(4)
	88.3(8)	25.8(11)	13.0(5)	6.8(5)
	93.6(7)	25.8(12)	12.5(7)	6.3(5)
	$98.6^{+0.7}_{-0.6}$	25.4(10)	11.6(6)	5.4(5)
	$79.0^{+2.0}_{-1.9}$	26.1(10)	13.7(6)	4.5(3)
	88.8(19)	24.6(14)	11.9(4)	4.7(3)
	97.8(16)	24.6(9)	11.6(8)	4.76(20)
	106.1(14)	24.6(9)	11.6(5)	3.90(18)
	114.1(13)	23.8(9)	10.5(4)	4.97(22)
	121.6(11)	23.8(9)	10.3(5)	4.76(20)
	$128.8^{+1.0}_{-0.9}$	23.7(9)	10.3(5)	4.8(3)
	135.8(8)	23.5(9)	9.5(10)	5.29(22)
	142.5(6)	23.3(9)	9.5(4)	5.20(24)
	149.0(5)	22.9(9)	9.0(4)	4.45(21)
	134.8(17)	23.2(11)	10.0(6)	4.9(3)
	141.5(16)	23.6(9)	9.5(5)	4.9(3)
	148.1(14)	22.2(8)	9.8(4)	3.81(22)
	154.3(13)	22.0(8)	9.6(4)	4.07(23)
	160.6(12)	21.9(8)	9.2(7)	3.73(18)
	166.5(10)	21.7(8)	9.2(5)	3.79(20)
	172.3(9)	22.1(10)	9.2(5)	4.9(3)
	178.1(7)	22.0(8)	8.5(4)	4.58(22)
	183.7(6)	21.8(8)	8.2(4)	5.0(3)
	189.1(4)	22.4(8)	7.4(4)	4.9(3)

B. Parametrization of the production cross sections

The data presented in this work have been fitted to Eq. 9. The range of the fit and its parameters are presented in the following table.

TABLE IX. Parameters of the production cross section fitting curve

Reaction Parameter	$^{12}\text{C}(\text{p},\text{x})^{11}\text{C}$	$^{16}\text{O}(\text{p},\text{x})^{11}\text{C}$	$^{14}\text{N}(\text{p},\text{x})^{11}\text{C}$	$^{16}\text{O}(\text{p},\text{x})^{13}\text{N}$	$^{14}\text{N}(\text{p},\text{x})^{13}\text{N}$	$^{16}\text{O}(\text{p},\text{x})^{15}\text{O}$
a ₀	86423.8	20837	27254	3302.52	1076.84	8471.57
a ₁	-4.53238e+06	-1.17523e+06	1.08172e+06	-344921	-440478	-166317
a ₂	1.18775e+08	2.83661e+07	-8.3163e+07	3.40013e+07	5.1973e+07	-277389
a ₃	-1.69095e+09	-3.73377e+08	4.79638e+08	-1.46591e+09	-1.13565e+09	3.02514e+06
a ₄	9.66632e+09	2.33149e+09	1.31897e+10	2.02267e+10	6.57578e+09	1.9191e+08
b ₀	2169.2	1265.09	1406.69	258.171	-377.248	52.5751
b ₁	-258391	-90285.8	-29509.3	1679.85	37082.4	-3007.49
b ₂	1.41445e+07	1.78423e+06	22058.5	-128921	109057	164369
b ₃	-3.70622e+08	712909	1733.01	-7.39076e+06	-4.35666e+07	-6.1241e+06
b ₄	3.73828e+09	5558.11	88.8574	3.88967e+08	7.49259e+08	8.58641e+07
Fit range	(14,190)	(19.5,190)	(21.8,190)	(19.5,190)	(9,190)	(16,190)